Bench exercise: mark it out before you cut

Japanese Joinery by Hand · Episode 6. Mark a full koshikake ari-tsugi on two pieces of softwood, following the episode's sequence, and check every line before any cutting. Cutting is optional: do it with a teacher if you can. About 60 minutes for the marking.

Safety. If you go on to cut: clamp the work, keep both hands behind the edge, and stop saw cuts at the line. Sharp-tool technique, and how to use these tools safely, is best learned from a teacher, at a bench, in person.

1 · Prepare

1. Two planed softwood blocks of the same square section (the episode uses 105 mm; 70–90 mm is fine), each about 400 mm long.
2. Decide which is the male (the upper piece) and the female, and write it on each.
3. Strike center lines on the top faces, as in Episode 1.

2 · Mark the female

1. On the top face: the end line, then the seat line, then the dovetail lines, laid out evenly on either side of the center line.
2. Angle the dovetail's sides using the width of your square's blade, running from the seat line toward the dovetail line.
3. Carry the seat lines down both side faces, at half the depth.

3 · Mark the male

1. On the top face: the end line, the dovetail's tip, and the cut lines.
2. In the episode's sequence, the end is cut square first, and only then are the center line and dovetail lines carried onto the fresh end grain. If you won't cut today, note where they will go.

4 · Check before cutting

- Is the dovetail on the male exactly the same size as the socket drawn on the female, measured from each center line?
- Does every line square around corners and meet itself?
- Can you say, for each cut, whether you will take, leave or split the line?

What you should have

- Two fully marked pieces you could hand to someone else to cut.

The sequence follows one training course (ものつくり大学 WBT). If you cut it: split the line on the seat cuts, taper the male's cheeks (leave the line at the top, take it at the bottom), and dig the socket's tip a little deeper than the line.